Predicting breast cancer grade from routine clinical data: an open multi-cohort study with transport to Saudi patients

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Abstract

Background: Histological grade guides breast cancer treatment but is assigned only after tissue examination and is moderately reproducible. It is unknown how much of grade the routine record carries, whether such models transport, and what they predict for Saudi patients. **Methods:** We combined 5025 graded invasive breast cancers from 7 public cohorts into one data model. Seven models, from a baseline to gradient boosting, were developed in METABRIC (2386 patients) with nested cross-validation and compared with logistic regression. Frozen models were validated in five external cohorts. We assessed discrimination, calibration and utility, measured leakage, tested 529 MRI radiomic features, and re-weighted the cohorts to published Saudi case-mix. **Results:** All learned models reached a macro-AUC of 0.72–0.73 internally; XGBoost did not beat logistic regression (difference +0.001, 95% CI -0.006 to +0.008). External macro-AUCs were 0.69–0.71 and 0.70–0.72 in the two full-feature cohorts. Calibration slopes stayed near one, but the models under-predicted grade 3 in one cohort and over-predicted it in another; an intercept update corrected both. Leakage shortcuts inflated macro-AUC by up to +0.297. Radiomics did not improve on clinical variables. Saudi case-mix raised each cohort's grade-3 share by +9.6 to +10.3 points, yet a METABRIC model would predict 59.1% grade 3 against 36.7% published. **Conclusions:** Routine variables predict grade moderately and consistently. Logistic regression is sufficient, but local recalibration is essential. Western-trained models would over-predict high-grade disease in Saudi patients.

Keywords: breast cancer, histological grade, clinical prediction model, external validation, calibration, data leakage, radiomics, transportability, Saudi Arabia, TRIPOD+AI

1. Introduction

Histological grade is one of the strongest prognostic factors in invasive breast cancer. The Nottingham (Elston–Ellis) system scores tubule formation, nuclear pleomorphism and mitotic count, and the resulting grade of 1, 2 or 3 predicts survival independently of tumour size and nodal status [1–3]. Grade has two practical weaknesses. First, it is only available after a pathologist has examined tissue. Second, it is only moderately reproducible: agreement between pathologists typically reaches a κ of 0.5–0.7 [4, 5], grading rates differ between laboratories in national data [6], and the large intermediate group, grade 2, is prognostically mixed [7, 8].

Every patient record, however, already contains variables that are linked to grade: age, menopausal status, oestrogen receptor (ER), progesterone receptor (PR) and HER2 status, histological type, tumour size and nodal status. If these variables predict grade reliably, a simple model could fill gaps in registry and trial data, flag records whose assigned grade is unlikely given the rest of the record, and provide the baseline that image-based grading tools should beat. Recent work has moved straight to images, predicting grade from whole-slide images with deep learning and pathology foundation models [9–11] and from MRI radiomics [12, 13]. We found no study that measured, on open data from several countries, how much grade the routine record already carries.

Problem statement. Three gaps prevent a reliable answer.

1. *No transparent baseline.* Published grade-prediction studies are mostly single-centre, rarely validate their models externally, and seldom report calibration, which is the property that most often fails when a model moves to a new hospital [14, 15]. It is also unclear whether flexible machine-learning methods add anything over regression, which systematic reviews suggest they usually do not [16].

2. *Hidden leakage.* Public breast cancer datasets include variables that contain grade by construction, such as the Nottingham Prognostic Index and the three Nottingham component scores. Using them as predictors, or fitting preprocessing steps before splitting the data, lets information about the outcome reach the model and inflates performance [17–19]. The size of this inflation for grade prediction has not been measured.

3. *Unknown transport to Saudi patients.* Breast cancer is the most common cancer in Saudi women; its age-standardised incidence rose from 11.8 to 29.7 per 100,000 between 2001 and 2017, when it accounted for 30.9% of female cancers [20]. Patients in Arab countries are younger and more often have hormone-receptor-negative or HER2-positive disease than patients in Western

cohorts [21–24]. No grade-labelled patient-level dataset from the region is openly available, and local studies are based on single-centre records that cannot be shared [25, 26]. It is therefore unknown whether models built on Western data would give correct predictions for Saudi patients [27, 28].

Research questions. We address these gaps with openly downloadable data only, so that every result can be reproduced by any reader. We ask: **RQ1**, how strongly is grade associated with each routine variable? **RQ2**, how well do models of increasing flexibility predict grade, within the development cohort and in independent cohorts from other countries, and are they well calibrated? **RQ3**, how much do common analytic shortcuts inflate apparent performance? **RQ4**, do MRI radiomic features add to the routine record? **RQ5**, what would Western-developed models predict for patients with a Saudi case-mix?

Contributions.

1. A harmonised, checksummed data model of 5025 grade-labelled invasive breast cancers from 7 public cohorts on three continents, built with explicit rules that exclude in-situ disease, keep lobular and ductal carcinoma separate and allow one record per patient (Figure 1).
2. A model comparison that follows TRIPOD+AI [29]: seven models from a prevalence baseline to gradient boosting, nested cross-validation, multiple imputation, paired comparisons with logistic regression, validation in five external cohorts and across datasets, calibration with model updating, and decision curves.
3. To our knowledge, the first measurement, on the same data and models, of how much four common leakage shortcuts inflate apparent performance for grade prediction.
4. A head-to-head test of 529 MRI radiomic features against the routine record in the same patients.
5. A transport analysis that re-weights open cohorts to published Saudi hospital series and shows what Western-developed models would imply for Saudi patients.

2. Methods

2.1. Study design and reporting

This is a secondary analysis of de-identified, publicly released data. We report the prediction analysis according to TRIPOD+AI [29] and the descriptive analysis according to STROBE [30]; the completed TRIPOD+AI checklist is in Supplementary Table S9. Figure 1 shows the study framework in three lanes: data

preparation (A), model development and validation (B), and the checks that test the results (C). Before running any experiment, we fixed the feature sets, models, metrics, expected directions of effect and re-weighting targets in a configuration file (random seed 20261017). All tables, figures and numbers in this paper are generated directly from the results files by code.

2.2. Data sources

All data were downloaded without registration from the cBioPortal data hub [31, 32], The Cancer Imaging Archive (TCIA) and the UCI Machine Learning Repository. A download script records a SHA-256 checksum for every file, so any later change by a data custodian can be detected. The cohorts and their roles were:

- **METABRIC** (UK/Canada; development) [33, 34]: a surgical cohort with Nottingham grade, tumour size in millimetres, number of positive nodes, receptor status and menopausal status.
- **MSK-2018** (USA; external validation, full feature set) [35]: patients sequenced at Memorial Sloan Kettering, enriched for advanced hormone-receptor-positive disease, with grade, T and N category, receptors and menopausal status at diagnosis.
- **Duke-Breast-Cancer-MRI** (USA; external validation, full feature set; imaging extension) [36]: 922 patients with invasive cancer imaged at Duke University Hospital between 2000 and 2014, with Nottingham grade and its component scores, receptors, T and N category, menopausal status and 529 pre-computed MRI features (CC BY-NC 4.0; obtained from a checksummed public copy of the TCIA files).
- **MBCproject** (USA/Canada; external, core features) [37, 38]: a patient-partnered cohort of metastatic disease with diagnosis-time records abstracted from charts; age is given in five-year bands.
- **Sanger** (UK/Europe; external, core features) [39]: 100 primary tumours with grade and the three Nottingham component scores.
- **Broad** (Mexico and Vietnam; external, core features) [40]: the only open grade-labelled cohort from low- and middle-income countries that we found.
- **BCCRC** (Canada; triple-negative; descriptive only) [41] and **TCGA-BRCA** (USA; descriptive only, no structured grade) [42].
- **Ljubljana** (Slovenia; sensitivity analysis) [43]: 286 patients with a "degree of malignancy" of 1–3 and grouped covariates.

We found no openly available, grade-labelled patient-level cohort from Saudi Arabia or the Gulf. Saudi data therefore enter the study only as published summary figures (Section 2.12).

2.3. Harmonisation to a common data model

We mapped every source to one row per patient with the following variables: age in years, post-menopausal status, ER, PR and HER2 status (positive or negative; equivocal results treated as missing), histological type (ductal/no special type, lobular, mixed ductal–lobular, other special type), T category (T1 $\leq$20 mm, T2 21–50 mm, T3 or T4) and nodal status (any positive regional node, including micrometastasis, versus none). Four rules prevented common data-handling errors: (i) the outcome is invasive Nottingham grade 1–3, and any other value is treated as missing; (ii) pure in-situ disease (DCIS, LCIS) and non-carcinomas are excluded, never relabelled as invasive; (iii) lobular carcinoma is never merged with ductal carcinoma; and (iv) men are excluded. When a patient had several sequenced samples (MSK-2018), we kept the diagnosis-time record once. Supplementary Table S1 shows the participant flow.

2.4. Outcome, predictors and feature sets

The primary outcome was grade in three classes; the main secondary outcome was grade 3 versus grades 1–2. We used two pre-specified feature sets. The *full* set (age, menopausal status, ER, PR, HER2, histology, T category and nodal status) was available in METABRIC, MSK-2018 and Duke. The *core* set (age, ER, PR, HER2 and histology) was available in all external cohorts. Variables that contain grade by construction—the Nottingham Prognostic Index and the three Nottingham component scores—and stage were never used as predictors; they appear only in the leakage audit. A unit test stops the analysis if any of them enters a feature set.

2.5. Descriptive and association statistics

We matched each association measure to the type of variable: Kendall's τ_b for grade against binary or ordinal variables, bias-corrected Cramér's V [44] for histological type, and Spearman's ρ with the Kruskal–Wallis ε^2 for age. Every estimate has a bootstrap 95% confidence interval (2000 resamples), and p values were adjusted for the 22 tests with the Benjamini–Hochberg procedure [45]. These associations describe which tumours tend to be high grade; they do not imply causation. A proportional-odds (ordinal logistic) model fitted to complete cases in METABRIC summarises the associations jointly.

2.6. Model development

We compared seven models of increasing flexibility: a baseline that predicts the class prevalences, multinomial logistic regression (LR) with an L_2 penalty, proportional-odds logistic regression, k-nearest neighbours, random forest, XGBoost [46] and LightGBM [47]. We also pre-specified a tabular foundation model, TabPFN [48], as an optional eighth model (status: not run; see Limitations).

Imputation of missing values [49], standardisation and the model itself were combined in one pipeline, so every step was fitted on training data only. We tuned hyper-parameters by grid search, minimising log-loss in an inner 3-fold cross-validation. We did not rebalance the classes, because resampling distorts predicted probabilities [50]. Instead, when a single predicted class was needed, we chose the class with the largest ratio of predicted probability to training prevalence, $\hat{y} = \arg\max_k p_k/\pi_k$.

Missing data. The primary analysis used single chained-equation imputation fitted within each training fold. We ran two pre-specified sensitivity analyses: multiple imputation with $m = 20$ imputed datasets, a separate model for each and the average of their predicted probabilities [51, 52]; and a complete-case analysis. Both used the same resampling design as the primary analysis.

Sample size. We calculated the minimum development sample size with the criteria of Riley et al. [53], applied to each pair of grades as recommended for multinomial models [54]. The criteria require a shrinkage factor of at least 0.9, a small difference between apparent and adjusted R^2 (δ=0.05) for every pairwise sub-model with 10 predictor parameters, and precise estimation of each grade's prevalence (±0.05). The expected pairwise R^2 values came from out-of-fold LR predictions, which are not optimistic.

2.7. Validation

Internal validation used nested, stratified 5-fold cross-validation in METABRIC, repeated 10 times. We averaged each patient's out-of-fold predictions across repeats and computed bootstrap 95% confidence intervals (2000 resamples) on these averages. Each model was compared with LR by a paired bootstrap of the difference in macro-AUC, using the same resamples for both models. For external validation, each model was tuned on all of METABRIC, frozen, and applied unchanged to MSK-2018 and Duke (full set) and to MSK-2018, Duke, MBCproject, Sanger and Broad (core set). We also used leave-one-dataset-out validation: models were trained on all but one of METABRIC, MSK-2018, Duke, MBCproject and Sanger and tested on the remaining cohort [55].

2.8. Performance measures

We measured discrimination with the macro-averaged one-versus-rest AUC and the AUC for each grade; classification with macro-F1, balanced accuracy and quadratic-weighted κ (QWK); and the overall accuracy of the predicted probabilities with the multiclass Brier score. For grade 3 versus 1–2 we also report the calibration intercept (whether predictions are too high or too low on average), the calibration slope (whether they are too extreme or too modest), smoothed calibration curves [14]

and decision curves [56]. For all three grades we compare the observed and mean predicted prevalence [57]. In each external cohort we tested a simple model update: re-estimating only the intercept, cross-fitted over two halves of the cohort. This separates miscalibration caused by a different grade prevalence from miscalibration of the predictor effects [15].

2.9. Leakage audit

We compared the correct protocol with four common shortcuts on the same patients and models [17]: (a) adding the Nottingham Prognostic Index, which includes grade, as a predictor; (b) imputing and oversampling with SMOTE [58] on the whole dataset before cross-validation; (c) adding the Nottingham component scores as predictors (Sanger and Duke); and (d) splitting by record instead of by patient when some patients have several records (MSK-2018). The last shortcut is the tabular version of splitting image tiles rather than patients in digital pathology [59].

2.10. Interpretability and clinical plausibility

We computed TreeSHAP values for the grade-3 output of XGBoost [60, 61]. Before the analysis we wrote down the expected direction of effect for seven features: ER-negative, PR-negative, HER2-positive, larger tumours, node-positive disease, younger age and non-lobular histology should all be associated with higher grade. A direction was counted as confirmed when the bootstrap interval of the Spearman correlation between a feature's value and its SHAP value excluded zero in the expected direction.

2.11. Imaging extension: MRI radiomics in Duke

In the graded Duke patients we compared three penalised multinomial LR models using nested cross-validation (10 repeats of 5 folds): the clinical full set, the 529 pre-computed dynamic contrast-enhanced MRI features [36], and both combined. The penalty was tuned in an inner 3-fold loop. The value added by radiomics was the paired-bootstrap difference in macro-AUC compared with the clinical model.

2.12. Transport to Saudi case-mix

Without Saudi patient data, we asked a narrower question: what grade distribution, and what model predictions, would the Western cohorts show if their patients had the receptor, histology and age profile reported for Saudi patients? We re-weighted the METABRIC, MSK-2018 and Duke patients by iterative proportional fitting (raking) so that their weighted profile matched published Saudi figures [28]. The primary target was a series of Saudi nationals treated at King Fahd University Hospital, Dammam [25], which reports ER, PR, HER2 and histology.

Because it reports only the median age (51 years), we assumed that 50% of patients were aged ≤50 years and varied this from 40% to 60% in a sensitivity analysis. The second target was a tertiary-centre series from King Fahad Medical City, Riyadh [26], which reports ER, PR, HER2, histology and the share aged ≤50 years; we recalculated its receptor percentages from the published counts. Every value was transcribed from the published tables, with table and page numbers recorded in the repository. For each target and cohort we report the re-weighted grade distribution with a bootstrap 95% confidence interval (500 resamples, each re-weighted afresh), the effective sample size of the weights, and the weighted mean predicted probabilities. We pooled the published Saudi grade-3 proportions with a DerSimonian–Laird random-effects model on the logit scale, using every series that reports grade for invasive cancer only [25, 26, 62]; a sensitivity analysis also included a small series that grades all lesions [63]. This analysis shows how much of the difference in grade can be explained by measured case-mix; it cannot capture local grading practice or unmeasured differences.

2.13. Software and reproducibility

We used Python 3.13 with scikit-learn 1.9.1, imbalanced-learn 0.14.2, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1 and statsmodels 0.15.0 (versions are pinned in the repository). A single command (`make all`) downloads and checks the data, runs the unit tests that enforce the harmonisation and leakage rules, runs experiments E1–E10 and regenerates every table, figure and number in the paper.

3. Results

3.1. Cohorts

After exclusions we analysed 5025 patients with a recorded grade in 7 cohorts, plus 1072 ungraded TCGA patients for description (Table 1; participant flow in Supplementary Table S1). The cohorts differed widely (Figure 2). Median age ranged from 43 years in MBCproject to 61 years in METABRIC. Grade 3 made up 50.2% of METABRIC, 67.6% of MSK-2018, which is enriched for advanced disease, and 32.5% of Duke. Compared with every open cohort except the triple-negative BCCRC series, the Dammam Saudi series [25] had fewer ER-positive (57.3%) and PR-positive (45.0%) tumours; only Sanger had more HER2-positive tumours (Dammam 25.3%). Even so, the Dammam grade distribution (8.7/54.5/36.7%) had more grade 2 and less grade 3 than METABRIC.

3.2. Associations between grade and routine variables

Of the 22 pre-specified associations, 20 remained significant after adjustment for multiple testing (Table 2, Figure 3). In METABRIC, high grade was most strongly

Table 1: Characteristics of the open cohorts after harmonisation and of two published Saudi hospital series. Percentages are of patients with the variable recorded (missingness in `results/e1_cohorts.csv`). MSK = MSK-2018; MBCP = MBCproject; Broad = Mexico and Vietnam; dev/ext/desc = development, external validation, descriptive. †Saudi nationals, King Fahd University Hospital, Dammam, 1998–2017 [25]. ‡King Fahad Medical City, Riyadh, 2016–2019 (90.1% Saudi nationals; 39.8% stage IV) [26]; receptor percentages recomputed from the published counts.

Role	METABRIC dev	MSK ext	Duke ext	MBCP ext	Sanger ext	Broad ext	BCCRC desc	TCGA desc	Dammam† anchor	Riyadh‡ anchor
Patients analysed, *n*	2386	1573	637	207	100	59	63	1072	286	283
Age, median (IQR), y	61 (51–70)	52 (43–61)	52 (45–61)	43 (38–53)	55 (44–66)	52 (42–58)	56 (48–63)	58 (49–67)	51	–
Age ≤ 50 y, %	23.4	45.7	42.4	70.0	43.0	45.8	33.3	30.4	–	45.2
Grade 1, %	8.9	5.7	17.7	12.1	8.0	6.8	6.3	–	8.7	4.2
Grade 2, %	40.9	26.7	49.8	50.2	41.0	49.2	4.8	–	54.5	47.0
Grade 3, %	50.2	67.6	32.5	37.7	51.0	44.1	88.9	–	36.7	48.8
IDC / NST, %	76.7	81.8	89.0	83.0	79.2	96.6	92.1	77.2	91.3	–
ILC, %	7.9	11.7	9.9	10.2	6.2	0.0	0.0	20.2	7.0	–
Mixed ductal–lobular, %	11.2	4.9	0.0	5.8	9.4	1.7	1.6	0.0	–	–
Other special type, %	4.2	1.5	1.1	1.0	5.2	1.7	6.3	2.6	1.7	–
ER positive, %	73.9	84.6	75.5	84.1	79.0	68.6	4.8	–	57.3	62.6
PR positive, %	52.1	72.2	64.8	74.7	60.0	58.8	1.6	–	45.0	54.1
HER2 positive, %	12.5	11.5	17.6	23.6	30.0	17.4	0.0	–	25.3	36.1
T1 (≤20 mm), %	43.8	49.2	44.9	–	–	–	42.6	25.8	–	–
T2 (21–50 mm), %	51.0	34.4	42.7	–	–	–	49.2	57.8	–	–
≥T3, %	5.2	16.3	12.4	–	–	–	8.2	16.4	–	–
Node positive, %	47.6	50.3	40.3	–	–	–	49.1	51.7	–	–
Post-menopausal, %	78.0	48.0	53.8	28.1	–	30.8	–	–	–	–

linked to ER-negative (τ_b -0.37) and PR-negative (τ_b -0.31) disease, followed by HER2 positivity (+0.19), larger tumours (+0.15), nodal involvement (+0.14) and younger age (ρ -0.12). Histological type showed a moderate association (V = 0.18). All directions were the same in MSK-2018. In the multivariable proportional-odds model (1800 complete cases; Table 5), ER positivity (odds ratio for higher grade 0.17, 95% CI 0.12–0.24) and lobular histology (0.41, 0.28–0.57) remained the strongest independent factors.

Table 2: Scale-appropriate associations with bootstrap 95% CIs and Benjamini–Hochberg-adjusted p values. Binary variables are coded 1 = present/positive, so a negative τ_b for ER means ER-positive tumours tend to be lower grade.

Pair	Measure	METABRIC		MSK-2018	
		Effect (95% CI)	p_{BH}	Effect (95% CI)	p_{BH}
Grade × ER+	τ_b	-0.373 (-0.404, -0.342)	<0.001	-0.211 (-0.241, -0.178)	<0.001
Grade × PR+	τ_b	-0.310 (-0.350, -0.271)	<0.001	-0.203 (-0.245, -0.165)	<0.001
Grade × HER2+	τ_b	+0.194 (+0.155, +0.229)	<0.001	+0.143 (+0.106, +0.181)	<0.001
Grade × T category	τ_b	+0.153 (+0.115, +0.191)	<0.001	+0.210 (+0.166, +0.257)	<0.001
Grade × Node+	τ_b	+0.137 (+0.097, +0.178)	<0.001	+0.172 (+0.117, +0.221)	<0.001
Grade × Post-menopausal	τ_b	-0.104 (-0.148, -0.061)	<0.001	-0.043 (-0.091, +0.004)	0.083
Age × Grade	ρ	-0.118 (-0.158, -0.078); ε^2=0.016	<0.001	-0.123 (-0.171, -0.074); ε^2=0.016	<0.001
Grade × Histology	V	+0.177 (+0.145, +0.217)	<0.001	+0.108 (+0.076, +0.157)	<0.001
Histology × ER+	V	+0.139 (+0.109, +0.174)	<0.001	+0.205 (+0.158, +0.263)	<0.001
T category × Node+	τ_b	+0.218 (+0.179, +0.256)	<0.001	+0.374 (+0.330, +0.417)	<0.001
Age × ER+	ρ	+0.267 (+0.227, +0.304); ε^2=0.071	<0.001	+0.017 (-0.033, +0.065); ε^2=0.000	0.488

3.3. *Internal validation: flexible models did not beat regression*

All learned models performed well above the baseline, and they performed almost identically (Table 3, Figure 4). Multinomial LR (macro-AUC 0.725) and the proportional-odds model (0.725) matched XGBoost (0.723) and the other ensemble methods. The paired difference between XGBoost and LR was +0.001 (95% CI -0.006 to +0.008), and no model exceeded LR by more than 0.009 even at the upper confidence limit. k-nearest neighbours performed worst (0.715). Results were stable across repeats (standard deviation at most 0.003). Grade 3 was the easiest class to identify (AUC 0.752 for Ordinal LR (prop. odds)) and grade 2 the hardest. Macro-F1 was 0.465 for LR compared with 0.223 for the baseline. Multiple imputation changed macro-AUC by at most 0.003 (Supplementary Table S5), and a complete-case analysis (1800 patients) gave 0.729 for LR. The sample-size criteria were not met: 4477 patients would be needed, against 2386 available. The limiting factor was the grade 1 versus grade 2 comparison, which carries little predictive signal (Supplementary Table S6).

Table 3: Internal validation in METABRIC (nested, 10× repeated stratified 5-fold cross-validation; full feature set). Point estimates are means over repeats; 95% CIs are patient-level bootstraps (2000 resamples) of repeat-averaged out-of-fold predictions. Δ is the paired-bootstrap difference in macro-AUC versus multinomial LR. QWK = quadratic-weighted κ.

Model	Macro-AUC (95% CI)	Δ vs LR (95% CI)	AUC G1	AUC G2	AUC G3	Macro-F1 (95% CI)	QWK	Brier
Majority class	0.499 (0.478–0.514)	-0.228 (-0.255, -0.203)	0.497	0.500	0.499	0.223 (0.217–0.229)	0.000	0.573
Multinomial LR	0.725 (0.707–0.742)	ref.	0.741	0.681	0.752	0.465 (0.445–0.486)	0.356	0.490
Ordinal LR (prop. odds)	0.725 (0.709–0.743)	+0.001 (-0.004, +0.006)	0.744	0.679	0.752	0.458 (0.435–0.476)	0.366	0.491
k-NN	0.715 (0.699–0.735)	-0.008 (-0.016, +0.001)	0.729	0.676	0.741	0.452 (0.427–0.468)	0.315	0.496
Random forest	0.724 (0.708–0.743)	+0.001 (-0.007, +0.009)	0.739	0.682	0.751	0.449 (0.424–0.463)	0.344	0.492
XGBoost	0.723 (0.708–0.743)	+0.001 (-0.006, +0.008)	0.737	0.678	0.753	0.458 (0.440–0.480)	0.344	0.492
LightGBM	0.719 (0.705–0.741)	-0.001 (-0.010, +0.007)	0.728	0.678	0.752	0.457 (0.443–0.484)	0.338	0.496

3.4. *External and leave-one-dataset-out validation*

When applied unchanged to new cohorts with the full feature set, the models reached macro-AUCs of 0.69–0.71 in MSK-2018 (LR 0.710, 95% CI 0.685–0.734) and 0.70–0.72 in Duke (LR 0.709, 0.681–0.737; 637 patients) (Table 4). This is close to the internal estimate, so discrimination transported well. With the core feature set, which lacks tumour size and nodal status, MSK-2018 discrimination fell to 0.64–0.67. The smaller cohorts reached up to 0.797, with wide confidence intervals. Leave-one-dataset-out estimates ranged from 0.659 to 0.778 (Supplementary Table S2). Of 28 paired external comparisons between a flexible model and LR, 1 favoured the flexible model and 1 favoured LR.

3.5. *Calibration, updating and clinical utility*

Predictions were well calibrated internally (calibration slope 1.00 for LR and 1.02 for XGBoost; Supplementary Table S3). In the external cohorts the slope

Table 4: External validation of models developed in METABRIC and applied without refitting. Calibration slope is for P(grade 3).

Target (features)	Model	n	Macro-AUC (95% CI)	G3-AUC (95% CI)	Macro-F1	Cal. slope (G3)
MSK-2018 (full)	Majority class	1573	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.269	–
MSK-2018 (full)	Multinomial LR	1573	0.710 (0.685–0.734)	0.727 (0.701–0.753)	0.372	1.03
MSK-2018 (full)	Ordinal LR (prop. odds)	1573	0.713 (0.688–0.737)	0.725 (0.699–0.750)	0.370	1.02
MSK-2018 (full)	Random forest	1573	0.704 (0.678–0.728)	0.725 (0.699–0.751)	0.350	1.11
MSK-2018 (full)	XGBoost	1573	0.711 (0.686–0.735)	0.726 (0.701–0.753)	0.373	1.03
MSK-2018 (full)	LightGBM	1573	0.705 (0.681–0.730)	0.720 (0.693–0.746)	0.388	0.82
Duke MRI (full)	Majority class	637	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.164	–
Duke MRI (full)	Multinomial LR	637	0.709 (0.681–0.737)	0.779 (0.739–0.816)	0.494	0.96
Duke MRI (full)	Ordinal LR (prop. odds)	637	0.705 (0.678–0.731)	0.778 (0.738–0.814)	0.487	0.92
Duke MRI (full)	Random forest	637	0.717 (0.689–0.744)	0.789 (0.752–0.826)	0.467	1.01
Duke MRI (full)	XGBoost	637	0.720 (0.691–0.748)	0.789 (0.750–0.826)	0.486	0.97
Duke MRI (full)	LightGBM	637	0.720 (0.691–0.748)	0.786 (0.749–0.824)	0.484	0.88
MSK-2018 (core)	Majority class	1573	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.269	–
MSK-2018 (core)	Multinomial LR	1573	0.667 (0.642–0.692)	0.685 (0.658–0.711)	0.349	0.90
MSK-2018 (core)	Ordinal LR (prop. odds)	1573	0.670 (0.645–0.695)	0.682 (0.655–0.709)	0.357	0.89
MSK-2018 (core)	Random forest	1573	0.665 (0.639–0.690)	0.690 (0.663–0.717)	0.357	0.98
MSK-2018 (core)	XGBoost	1573	0.658 (0.632–0.684)	0.680 (0.652–0.707)	0.357	0.88
MSK-2018 (core)	LightGBM	1573	0.660 (0.633–0.686)	0.681 (0.654–0.708)	0.357	0.72
MBCproject (core)	Majority class	207	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.182	–
MBCproject (core)	Multinomial LR	207	0.701 (0.647–0.754)	0.771 (0.703–0.832)	0.486	1.16
MBCproject (core)	Ordinal LR (prop. odds)	207	0.715 (0.665–0.761)	0.762 (0.692–0.829)	0.513	1.08
MBCproject (core)	Random forest	207	0.693 (0.640–0.744)	0.753 (0.686–0.819)	0.440	1.12
MBCproject (core)	XGBoost	207	0.697 (0.642–0.750)	0.758 (0.689–0.822)	0.455	1.01
MBCproject (core)	LightGBM	207	0.703 (0.649–0.757)	0.760 (0.693–0.826)	0.485	0.93
Sanger (core)	Majority class	100	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.225	–
Sanger (core)	Multinomial LR	100	0.797 (0.712–0.872)	0.806 (0.714–0.882)	0.570	1.33
Sanger (core)	Ordinal LR (prop. odds)	100	0.763 (0.679–0.842)	0.801 (0.704–0.882)	0.495	1.25
Sanger (core)	Random forest	100	0.764 (0.680–0.840)	0.772 (0.676–0.856)	0.441	1.27
Sanger (core)	XGBoost	100	0.766 (0.678–0.847)	0.774 (0.677–0.859)	0.479	1.20
Sanger (core)	LightGBM	100	0.767 (0.684–0.845)	0.774 (0.675–0.862)	0.486	1.03
Broad (MX/VN) (core)	Majority class	59	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.204	–
Broad (MX/VN) (core)	Multinomial LR	59	0.784 (0.691–0.869)	0.737 (0.599–0.863)	0.663	0.90
Broad (MX/VN) (core)	Ordinal LR (prop. odds)	59	0.777 (0.687–0.863)	0.738 (0.604–0.866)	0.675	0.86
Broad (MX/VN) (core)	Random forest	59	0.795 (0.678–0.891)	0.793 (0.653–0.910)	0.403	1.30
Broad (MX/VN) (core)	XGBoost	59	0.787 (0.668–0.882)	0.773 (0.634–0.888)	0.434	1.04
Broad (MX/VN) (core)	LightGBM	59	0.785 (0.665–0.887)	0.760 (0.616–0.884)	0.562	0.96
Duke MRI (core)	Majority class	637	0.500 (0.500–0.500)	0.500 (0.500–0.500)	0.164	–
Duke MRI (core)	Multinomial LR	637	0.702 (0.675–0.729)	0.783 (0.743–0.820)	0.434	1.01
Duke MRI (core)	Ordinal LR (prop. odds)	637	0.705 (0.678–0.732)	0.782 (0.743–0.819)	0.476	0.98
Duke MRI (core)	Random forest	637	0.715 (0.686–0.743)	0.782 (0.744–0.821)	0.485	1.02
Duke MRI (core)	XGBoost	637	0.714 (0.684–0.743)	0.784 (0.746–0.822)	0.476	0.97
Duke MRI (core)	LightGBM	637	0.712 (0.683–0.739)	0.779 (0.740–0.817)	0.487	0.88

stayed close to one (MSK-2018 1.03; Duke 0.96), which means the predictor effects transported. The average level of the predictions did not. In MSK-2018, 0.68 of patients had grade 3 but the mean predicted probability was only 0.47 (calibration intercept +0.99), so the model under-predicted grade 3. In Duke the opposite occurred: 0.32 observed against 0.52 predicted (intercept -1.02). Re-estimating only the intercept in each cohort removed this miscalibration without changing discrimination (Figure 5; per-grade results in Supplementary Table S4). Decision curves showed a net benefit over treating all or no patients as grade 3 for threshold probabilities above about 0.26 internally and 0.63 in MSK-2018, where the high prevalence of grade 3 makes "treat all" a strong alternative, and between 0.19 and 0.64 in Duke.

3.6. Interpretability and clinical plausibility

The SHAP analysis ranked ER positive as the most important feature and T category as the second (Figure 6). 7 of 7 pre-specified directions were confirmed, with bootstrap intervals excluding zero (Table 5), and all agreed in sign with the proportional-odds odds ratios. The model therefore uses the variables in the way a pathologist would expect.

Table 5: Clinical-plausibility check. "Expected" was pre-specified. Ordinal odds ratios are for higher grade from a complete-case proportional-odds model (bootstrap 95% CI).

Feature	mean|SHAP|	Direction ρ (95% CI)	Expected	Agrees	Ordinal OR (95% CI)
ER positive	0.330	-0.76 (-0.78, -0.74)	↓	yes	0.17 (0.12–0.24)
T category	0.158	+0.82 (+0.81, +0.83)	↑	yes	1.61 (1.31–2.00)
PR positive	0.135	-0.87 (-0.87, -0.86)	↓	yes	0.57 (0.46–0.72)
Node positive	0.086	+0.87 (+0.86, +0.87)	↑	yes	1.29 (1.08–1.62)
HER2 positive	0.065	+0.57 (+0.54, +0.60)	↑	yes	1.83 (1.28–2.69)
Mixed (vs IDC)	0.064	-0.55 (-0.57, -0.52)	–	–	0.53 (0.38–0.74)
Age (per 10 y)	0.061	-0.25 (-0.30, -0.21)	↓	yes	0.93 (0.83–1.03)
ILC (vs IDC)	0.053	-0.47 (-0.49, -0.43)	↓	yes	0.41 (0.28–0.57)
Other type (vs IDC)	0.034	-0.35 (-0.38, -0.31)	–	–	0.26 (0.16–0.43)
Post-menopausal	0.008	-0.65 (-0.68, -0.61)	–	–	0.96 (0.68–1.37)

3.7. Leakage audit

Every shortcut inflated apparent performance (Table 6, Figure 7). Adding the Nottingham Prognostic Index raised the macro-AUC of XGBoost from 0.721 to 0.972. Adding the Nottingham component scores raised LR from 0.696 to 0.962 in Sanger and from 0.698 to 0.995 in Duke. Imputing and oversampling before splitting raised k-nearest neighbours from 0.652 to 0.806. Splitting by record rather than by patient had a smaller effect (0.544 to 0.579 for 1-nearest neighbour) because few MSK-2018 patients had more than one record. The largest inflation,

+0.297 in macro-AUC, is larger than the entire gap between the baseline and the best correctly validated model.

Table 6: Leakage audit: identical models evaluated with the correct and a leaky protocol (5-fold cross-validation).

Scenario	Model	Honest AUC	Leaky AUC	Inflation
Target leakage: NPI as predictor (METABRIC)	Multinomial LR	0.726	0.949	+0.223
Target leakage: NPI as predictor (METABRIC)	XGBoost	0.721	0.972	+0.251
SMOTE + imputation before split (METABRIC)	k-NN	0.652	0.806	+0.154
SMOTE + imputation before split (METABRIC)	XGBoost	0.669	0.850	+0.181
Grade components as predictors (Sanger)	Multinomial LR	0.696	0.962	+0.266
Grade components as predictors (Duke MRI)	Multinomial LR	0.698	0.995	+0.297
Record-level split, 1718 records / 1573 patients (MSK)	k-NN	0.544	0.579	+0.035
Record-level split, 1718 records / 1573 patients (MSK)	Random forest	0.621	0.653	+0.033

3.8. Imaging extension: MRI radiomics in Duke

In the 637 graded Duke patients, the clinical model reached a cross-validated macro-AUC of 0.694 (95% CI 0.663–0.727). Radiomics alone reached 0.608 (0.575–0.645), and the combined model 0.648 (0.618–0.685) (Table 7, Figure 8). Adding radiomics to the clinical variables therefore lowered performance, by -0.045 (-0.077 to -0.011).

Table 7: Duke-Breast-Cancer-MRI: clinical, radiomics and fusion models (nested 10×5-fold cross-validation; L_2-penalised multinomial LR; paired bootstrap Δ versus the clinical model).

Model (L2 logistic)	Features	Macro-AUC (95% CI)	G3-AUC (95% CI)	Δ macro-AUC vs clinical (95% CI)	Brier
Clinical (full set)	10	0.694 (0.663–0.727)	0.769 (0.731–0.811)	ref.	0.551
DCE-MRI radiomics (529 features)	529	0.608 (0.575–0.645)	0.667 (0.626–0.718)	-0.085 (-0.126, -0.045)	0.600
Clinical + radiomics	539	0.648 (0.618–0.685)	0.731 (0.693–0.777)	-0.045 (-0.077, -0.011)	0.582

3.9. Transport to Saudi case-mix

When METABRIC was re-weighted to the Dammam receptor, histology and age profile (effective sample size 1088 of 1861), its grade distribution moved to 6.2/32.9/60.9% (grade 3 95% CI 58.5–63.0%), compared with 50.6% grade 3 observed in METABRIC and 36.7% published for Dammam (Table 8, Figure 9). In other words, the Saudi case-mix predicts *more* grade 3, not less. The same pattern

appeared for MSK-2018 (77.3%) and Duke (42.1%), for every assumed share of patients aged ≤50 years from 40% to 60% (grade 3 60.5–61.3%; Supplementary Table S7), and for the Riyadh series (re-weighted METABRIC grade 3 57.3%, 54.6–60.0%, against 48.8% published). Across 3 Saudi series (908 patients), the pooled published grade-3 share was 44.7% (37.2–52.5%); including the Najran series [63] gave 42.1% (34.5–50.0%). Under the Dammam case-mix, LR discrimination rose from 0.727 to 0.790, but the mean predicted probability of grade 3 was 59.1%, about 22 percentage points above the published Dammam figure.

Table 8: Re-weighting to published Saudi case-mix. ESS = effective sample size of the raking weights (first row of each block: patients with complete raking variables). Grade-3 CI: bootstrap with re-raking. Arrows: unweighted→Saudi-weighted LR discrimination. †Dammam series [25]; ‡Riyadh series [26] (tertiary referral centre). Pooled: DerSimonian–Laird random effects across 3 Saudi series reporting grade [25, 26, 62]; the Madinah series reports grade for invasive ductal carcinoma only and no receptor status, so it contributes to the pooled estimate but not to raking.

Anchor	Cohort	Quantity	G1	G2	G3 (95% CI)	ESS	Macro-AUC	G3-AUC
Dammam (Kussaibi 2021)†	METABRIC	observed	8.9	40.5	50.6	1861		
		re-weighted	6.2	32.9	60.9 (58.5–63.0)	1088		
		LR predicted (weighted)	6.7	34.2	59.1		0.727→0.790	0.759→0.811
		published	8.7	54.5	36.7			
Dammam (Kussaibi 2021)†	MSK-2018	observed	5.7	26.7	67.6	1496		
		re-weighted	3.4	19.2	77.3 (75.1–79.9)	846		
		LR predicted (weighted)	5.6	33.1	61.3		0.715→0.766	0.734→0.765
		published	8.7	54.5	36.7			
Dammam (Kussaibi 2021)†	Duke MRI	observed	17.7	49.8	32.5	637		
		re-weighted	13.2	44.7	42.1 (37.7–45.9)	510		
		LR predicted (weighted)	5.8	32.8	61.4		0.709→0.717	0.779→0.755
		published	8.7	54.5	36.7			
Riyadh (Durrani 2025)‡	METABRIC	observed	8.9	40.5	50.6	1861		
		re-weighted	8.2	34.5	57.3 (54.6–60.0)	852		
		LR predicted (weighted)	8.7	35.6	55.7		0.727→0.785	0.759→0.806
		published	4.2	47.0	48.8			
Riyadh (Durrani 2025)‡	MSK-2018	observed	5.7	26.7	67.6	1496		
		re-weighted	4.3	19.5	76.3 (73.8–78.6)	555		
		LR predicted (weighted)	6.8	34.8	58.4		0.715→0.775	0.734→0.771
		published	4.2	47.0	48.8			
Riyadh (Durrani 2025)‡	Duke MRI	observed	17.7	49.8	32.5	637		
		re-weighted	14.0	50.1	35.9 (32.5–39.7)	277		
		LR predicted (weighted)	8.3	34.5	57.1		0.709→0.710	0.779→0.776
		published	4.2	47.0	48.8			
Pooled Saudi anchors (random effects)		published	9.2	–	44.7 (37.2–52.5)			

3.10. Sensitivity analysis

In the Ljubljana cohort (286 patients, grouped covariates and no receptor data), the best macro-AUC for "degree of malignancy" was 0.636 (Supplementary Table S8). This confirms that discrimination drops when receptor status is not available.

4. Discussion

4.1. Principal findings

This study shows that the routine clinicopathological record carries moderate, reproducible information about breast cancer grade. Macro-AUCs were 0.72–0.73 internally and 0.64–0.80 across five external cohorts from three continents. Grade 3 was the easiest class to recognise and grade 2 the hardest, which mirrors the known difficulty of grade 2 for pathologists [4, 6]. Hormone-receptor status was the most informative variable, and tumour size, nodal status, HER2 and lobular histology contributed in the directions that clinicians would expect.

Three further findings matter for anyone who builds or uses such models. First, flexibility did not help: no flexible learner beat multinomial logistic regression internally, and external differences were rare and went in both directions. This agrees with systematic reviews of clinical prediction models [16] and is unsurprising when the predictors are few and mostly binary. Second, discrimination and predictor effects transported well (calibration slopes close to one), but the average level of the predictions did not. The same frozen model under-predicted grade 3 in MSK-2018 and over-predicted it in Duke, because the share of grade 3 tumours differs more between cohorts than their predictors do. A simple intercept update fixed the problem in both cohorts. Third, the size of the leakage effect was striking. Each shortcut that is easy to make with public breast cancer data inflated performance, by up to +0.297 in macro-AUC. That is more than the whole distance between a useless model and the best correctly validated one.

4.2. What the imaging extension adds

In the Duke cohort, 529 MRI radiomic features did not improve on the routine record, and adding them lowered cross-validated discrimination (paired Δ -0.045, -0.077 to -0.011). With 637 graded patients and hundreds of correlated features, the model could not find signal beyond what receptor status, tumour size and nodal status already provide. Published radiomics results for grade [12, 13] should therefore be judged against a clinical baseline evaluated in the same patients, as done here.

4.3. What this means for Saudi Arabia

Re-weighting to the Saudi profile raised the grade-3 share of every Western cohort by +9.6 to +10.3 percentage points. The Saudi case-mix—fewer receptor-positive and more HER2-positive tumours—is therefore associated with more high-grade disease. The level reached, however, depended on the source cohort (42.1–77.3% for the Dammam profile), because the Western cohorts already differed in their grade-3 share (32.5–67.6%). Published Saudi series report less grade 3

than the METABRIC- and MSK-based projections: 36.7% in Dammam, 48.8% in Riyadh and 44.7% when pooled across 3 series. A model developed in METABRIC would predict, on average, 59.1% grade 3 for patients with the Dammam profile, about 22 percentage points too many. Several explanations are possible: differences in grading practice (grade 2 is the least reproducible category), referral and era effects in hospital series, or real differences in tumour biology. Summary data cannot separate them. The practical message is clear, though. A Western grade model should not be used for Saudi patients without recalibrating its intercept, which needs only the local grade distribution. The most valuable next step is an openly shared, de-identified, grade-labelled Saudi cohort; the code released with this paper can validate and recalibrate the models on such a cohort with one additional data-mapping function.

4.4. Strengths and limitations

The main strengths are full reproducibility from public data with recorded checksums, an analysis plan fixed before the experiments, five external cohorts from three continents (one recruited through imaging rather than genomics), paired model comparisons, and an explicit measurement of leakage.

The study also has important limitations. Grade was assigned by different pathologists, in different eras, under slightly different conventions; the Ljubljana "degree of malignancy" predates the Elston–Ellis modification. Disagreement between pathologists [4, 5] limits the discrimination that any model can reach. Several cohorts were assembled for genomic studies or enriched for advanced disease, so they are not population-based. Tumour size was reduced to T category, and tumour size and nodal status were missing in three external cohorts. The sample-size criteria were not met: the grade 1 versus grade 2 comparison, which carries little signal, would need 4477 patients against 2386 available. Penalised estimation and calibration slopes close to one suggest that overfitting was small in practice, but estimates at the grade 1/2 boundary should be read with care. The Saudi analysis relies on published summary figures, not patient data. The Dammam series does not report the share of patients aged $\leq$50 years (the results were not sensitive to the assumed value); the Riyadh series comes from a referral centre with many stage IV patients and about 10% non-Saudi patients; the Madinah series grades ductal carcinoma only; and the Najran series grades all lesions, including in-situ and benign ones, so it was used only in a sensitivity analysis. The pre-specified tabular foundation model (TabPFN) could not be run because its weights are distributed through a gated repository that was not reachable from the build environment; the code path is in place (`make tabpfn`). Finally, the image-level analyses that we had planned with the Saudi KAU-BCMD mammograms [64] and the Egyptian CDD-CESM and BUSI collections [65, 66] could not be run, because

their hosts were not reachable from the build environment.

4.5. Implications for practice and research

A routine-variable model is not a substitute for histological grading. It has three realistic uses: imputing missing grade in registry or trial data with calibrated uncertainty; flagging records whose assigned grade is unlikely given the rest of the record, as a quality-assurance check; and serving as the minimum baseline that image-based grading tools must beat in the same patients, with patient-level data splits and with grade-encoding columns removed. Each use requires local recalibration of the intercept.

5. Conclusions

This study answers a simple question with open, reproducible evidence: routine clinicopathological variables predict breast cancer grade moderately and consistently, with macro-AUCs of about 0.73 internally and up to 0.80 in external cohorts from three continents.

The results contribute to knowledge in four ways. First, they set a transparent, externally validated benchmark for grade prediction that image-based methods can be measured against. Second, they show that logistic regression is enough for this task; gradient boosting, random forests and MRI radiomics added nothing. Third, they show that predictor effects transport across countries but baseline grade frequencies do not, so calibration, not discrimination, is the weak point when a grade model moves to a new hospital. Fourth, to our knowledge they are the first to quantify how much common leakage shortcuts inflate apparent performance in grade prediction—by up to +0.297 in macro-AUC—which gives reviewers and readers a concrete reason to distrust near-perfect results.

For practitioners, the message is practical. A simple, interpretable regression model can impute missing grade or flag implausible grades, provided it is recalibrated to the local grade distribution. Models that use the Nottingham Prognostic Index or the component scores, or that preprocess data before splitting, should not be trusted. For Saudi Arabia, the results show that the local case-mix implies more high-grade disease than in Western cohorts, yet Western-trained models would still over-predict grade 3 compared with published Saudi series. Local recalibration is therefore essential, and an openly shared, grade-labelled Saudi cohort would be the most valuable next contribution. All code, settings and results are public, so these findings can be checked, updated and extended as new data become available.

Data and code availability

All data are publicly available from their custodians: cBioPortal studies `brca_metabric`, `breast_msk_2018`, `brca_mbcproject_2022`, `brca_sanger`, `brca_broad`, `brca_bccrc` and `brca_tcga_pan_can_atlas_2018`; the TCIA Duke-Breast-Cancer-MRI clinical and imaging-feature tables (CC BY-NC 4.0; non-commercial use); and the UCI Breast Cancer dataset (doi:10.24432/C51P4M). Data are not redistributed: a download script retrieves them and verifies the SHA-256 checksums recorded in the repository (files retrieved 17–18 October 2026). Code, configuration, result files and the manuscript sources are available at https://anonymous.4open.science/r/bc-grade-opendata; `make all` reproduces every table, figure and in-text number.

Ethics statement

Secondary analysis of de-identified, publicly released data; no ethics approval or consent was required. The study was not registered.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude (Anthropic) to assist with analysis code, reference checking and drafting of the text. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. [Confirm or edit to reflect actual use.]

Declarations

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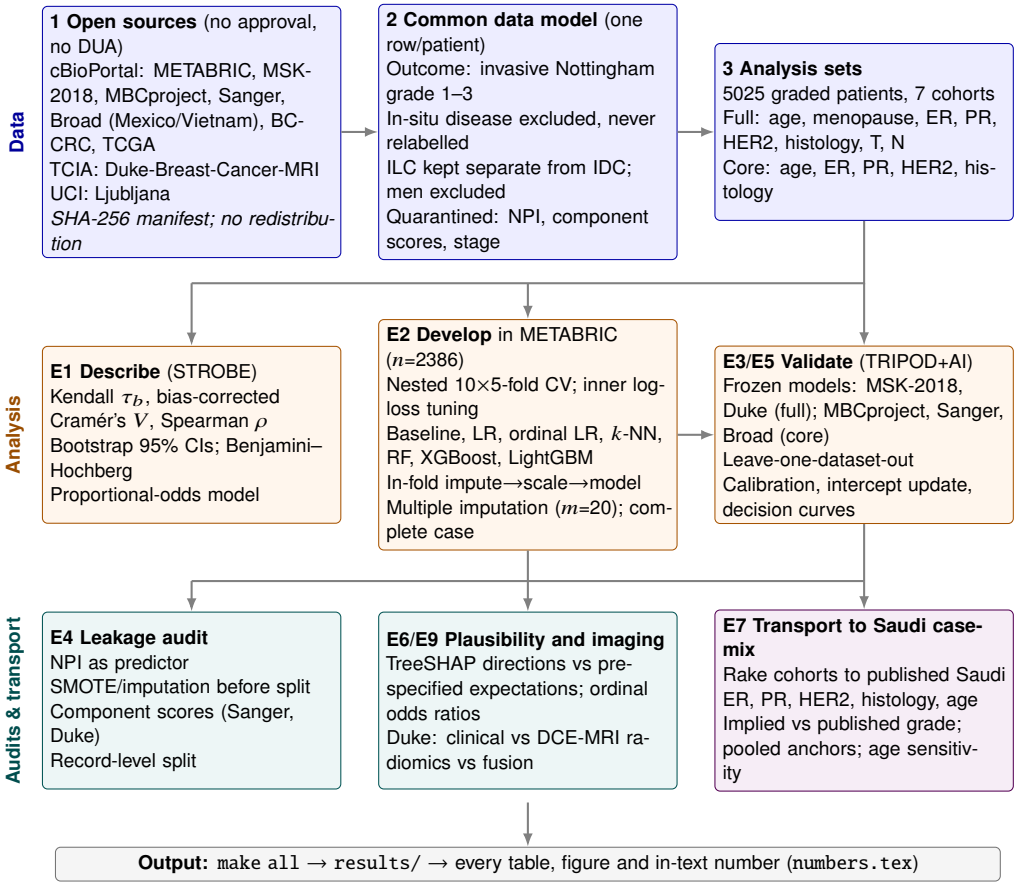


Figure 1: Study framework. **A**, open data are downloaded with checksums and mapped to one row per patient under four harmonisation rules; variables that contain grade are set aside. **B**, associations are measured with statistics suited to each variable type; models are developed in METABRIC with nested cross-validation, then frozen and validated in five external cohorts and across datasets. **C**, the results are checked for leakage and clinical plausibility, extended with MRI radiomics in the Duke cohort, and transported to published Saudi case-mix. E1–E10 are the experiment stages in the code repository.

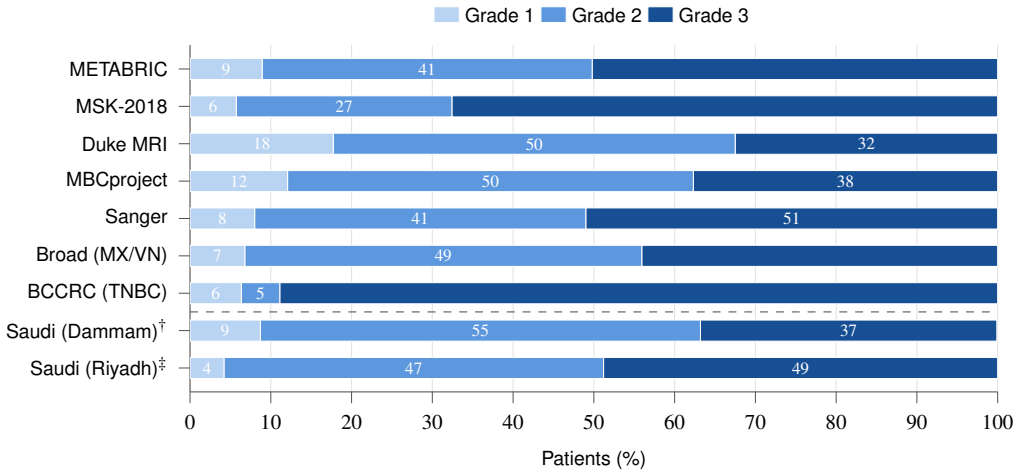


Figure 2: Distribution of Nottingham grade by cohort (%), with the published Saudi series below the dashed line (†25; ‡26).

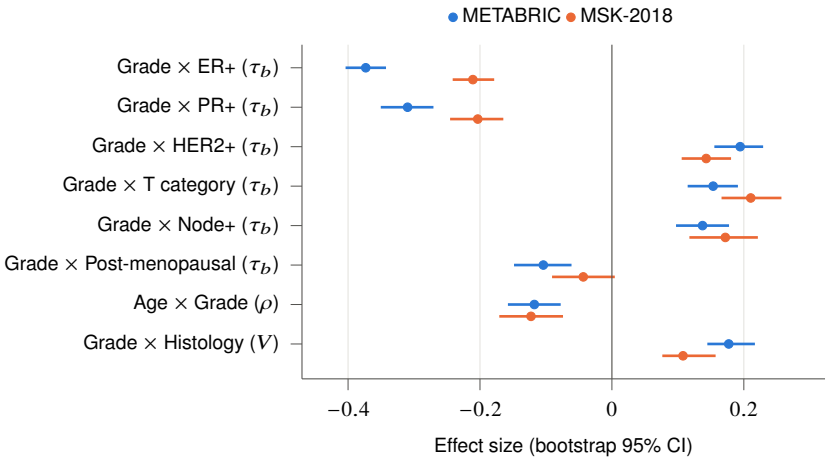


Figure 3: Associations involving grade in the two large cohorts (effect and bootstrap 95% CI).

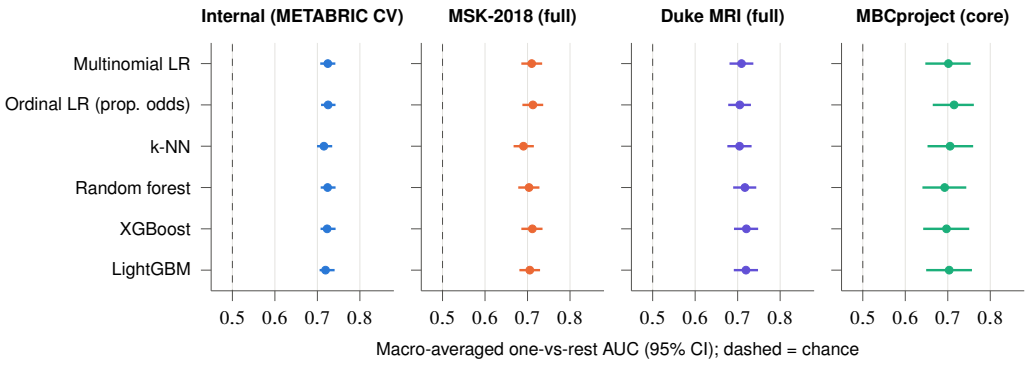


Figure 4: Macro-averaged AUC (95% CI) of the model ladder internally and in three external cohorts; dashed line = chance.

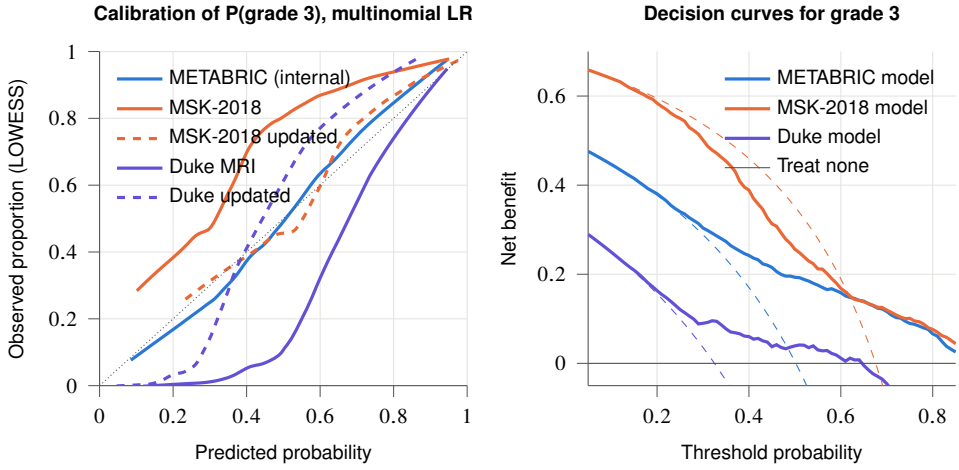


Figure 5: Left: LOWESS calibration curves for P(grade 3) from multinomial LR, internally, externally (solid) and after a cross-fitted intercept update (dashed). Right: decision curves (solid = model; thin dashed = treat all, same colour; treat none = 0).

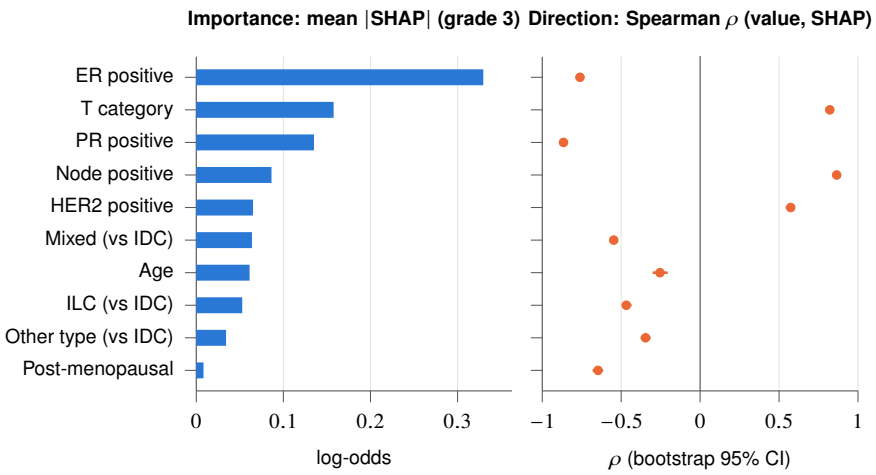


Figure 6: Global importance (mean |SHAP|) and direction (Spearman ρ between feature value and SHAP value, bootstrap 95% CI) for the grade-3 output of XGBoost in METABRIC. Beeswarm in Supplementary Figure S1.

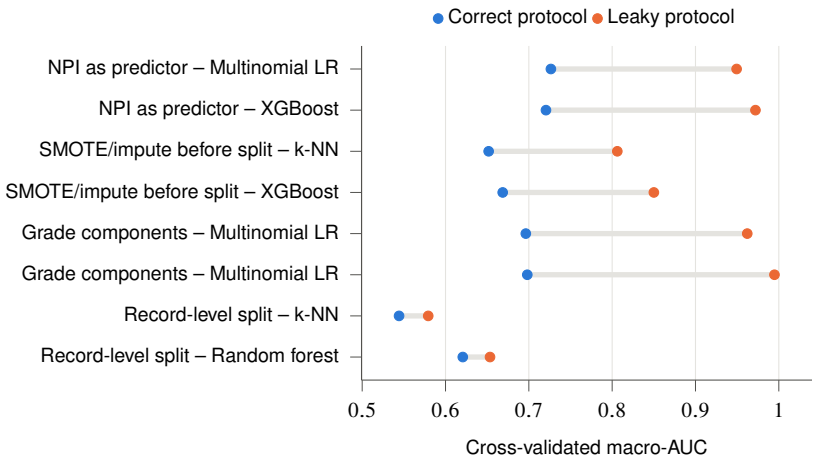


Figure 7: Apparent macro-AUC under the correct (blue) and leaky (orange) protocols.

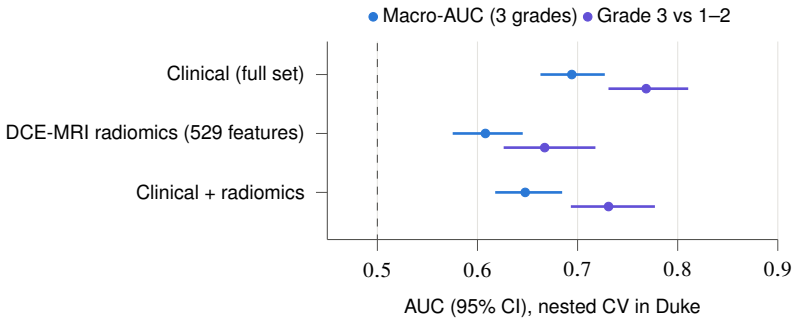


Figure 8: Duke: discrimination of clinical, radiomics and fusion models (95% CI); dashed = chance.

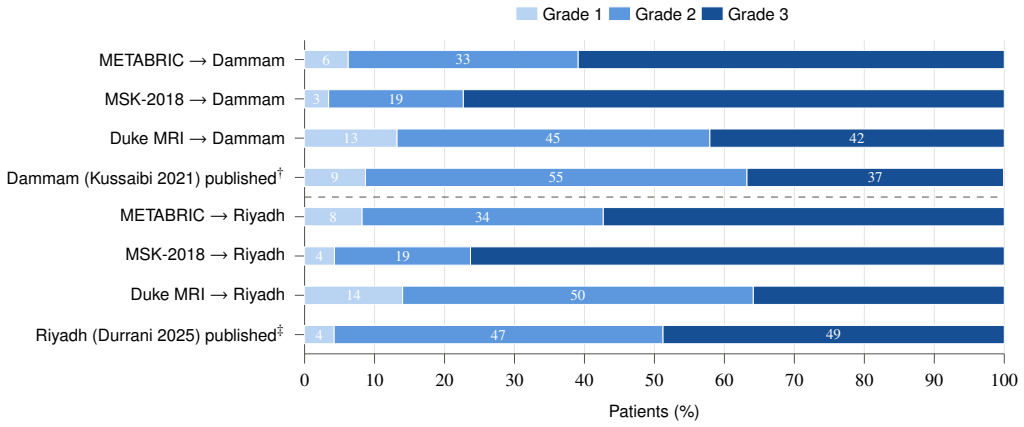


Figure 9: Grade distributions implied by re-weighting the development and external cohorts to each Saudi anchor's case-mix, and the published Saudi distributions.

Supplementary material

Table S1: Participant flow (STROBE).

Cohort	Step	*n*
METABRIC	records	2509
METABRIC	invasive carcinoma, female (excl. stage 0 / sarcoma)	2484
METABRIC	analysis set (grade 1-3 recorded)	2386
MSK-2018	samples	1918
MSK-2018	patients	1756
MSK-2018	female	1746
MSK-2018	analysis set (grade 1-3 recorded)	1573
MBCproject	patients	301
MBCproject	female, abstraction complete	256
MBCproject	invasive carcinoma	254
MBCproject	analysis set (grade 1-3 recorded)	207
Sanger	patients	100
Sanger	analysis set (grade 1-3 recorded)	100
Broad (MX/VN)	patients	103
Broad (MX/VN)	invasive carcinoma	94
Broad (MX/VN)	analysis set (grade 1-3 recorded)	59
BCCRC (TNBC)	patients	65
BCCRC (TNBC)	carcinoma	64
BCCRC (TNBC)	analysis set (grade 1-3 recorded)	63
Duke MRI	patients	922
Duke MRI	invasive carcinoma	921
Duke MRI	analysis set (grade 1-3 recorded)	637
TCGA-BRCA	patients	1084
TCGA-BRCA	female	1072

Table S2: Leave-one-dataset-out validation (core feature set).

Held-out cohort	Model	n	Macro-AUC (95% CI)	G3-AUC (95% CI)	Macro-F1
METABRIC	Multinomial LR	2386	0.698 (0.679–0.716)	0.727 (0.706–0.747)	0.393
METABRIC	Ordinal LR (prop. odds)	2386	0.696 (0.678–0.713)	0.728 (0.707–0.748)	0.413
METABRIC	Random forest	2386	0.702 (0.683–0.720)	0.730 (0.709–0.749)	0.410
METABRIC	XGBoost	2386	0.700 (0.681–0.718)	0.725 (0.704–0.745)	0.412
METABRIC	LightGBM	2386	0.689 (0.671–0.706)	0.724 (0.704–0.745)	0.433
MSK-2018	Multinomial LR	1573	0.659 (0.632–0.685)	0.684 (0.657–0.711)	0.285
MSK-2018	Ordinal LR (prop. odds)	1573	0.670 (0.645–0.695)	0.681 (0.654–0.709)	0.257
MSK-2018	Random forest	1573	0.664 (0.638–0.691)	0.676 (0.649–0.703)	0.302
MSK-2018	XGBoost	1573	0.662 (0.635–0.689)	0.672 (0.645–0.700)	0.315
MSK-2018	LightGBM	1573	0.661 (0.634–0.688)	0.673 (0.645–0.700)	0.335
MBCproject	Multinomial LR	207	0.709 (0.656–0.759)	0.778 (0.713–0.840)	0.322
MBCproject	Ordinal LR (prop. odds)	207	0.716 (0.665–0.763)	0.771 (0.706–0.835)	0.483
MBCproject	Random forest	207	0.707 (0.658–0.756)	0.770 (0.703–0.835)	0.332
MBCproject	XGBoost	207	0.710 (0.658–0.760)	0.773 (0.705–0.839)	0.358
MBCproject	LightGBM	207	0.714 (0.664–0.764)	0.764 (0.697–0.831)	0.329
Sanger	Multinomial LR	100	0.778 (0.692–0.855)	0.787 (0.692–0.871)	0.483
Sanger	Ordinal LR (prop. odds)	100	0.759 (0.675–0.838)	0.791 (0.696–0.871)	0.464
Sanger	Random forest	100	0.769 (0.683–0.847)	0.786 (0.690–0.869)	0.391
Sanger	XGBoost	100	0.774 (0.683–0.857)	0.791 (0.695–0.875)	0.442
Sanger	LightGBM	100	0.761 (0.670–0.845)	0.791 (0.695–0.874)	0.455
Duke MRI	Multinomial LR	637	0.703 (0.676–0.730)	0.782 (0.743–0.820)	0.459
Duke MRI	Ordinal LR (prop. odds)	637	0.705 (0.678–0.732)	0.782 (0.744–0.820)	0.465
Duke MRI	Random forest	637	0.717 (0.688–0.744)	0.787 (0.749–0.825)	0.460
Duke MRI	XGBoost	637	0.712 (0.682–0.739)	0.782 (0.743–0.822)	0.446
Duke MRI	LightGBM	637	0.708 (0.678–0.736)	0.780 (0.741–0.819)	0.458

Table S3: Calibration of P(grade 3): intercept (calibration-in-the-large) and slope, internally, in the external full-feature cohorts and after a cross-fitted intercept update.

Model	Setting	n	Obs. G3	Mean pred.	Intercept	Slope	G3-AUC
Multinomial LR	METABRIC (internal CV)	2386	0.502	0.502	-0.00	1.00	0.753
Multinomial LR	MSK-2018 (external)	1573	0.676	0.472	+0.99	1.03	0.727
Multinomial LR	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.03	0.727
Multinomial LR	Duke MRI (external)	637	0.325	0.517	-1.02	0.96	0.779
Multinomial LR	Duke MRI (intercept-updated)	637	0.325	0.325	+0.00	0.96	0.779
Ordinal LR (prop. odds)	METABRIC (internal CV)	2386	0.502	0.501	+0.00	0.96	0.752
Ordinal LR (prop. odds)	MSK-2018 (external)	1573	0.676	0.474	+0.97	1.02	0.725
Ordinal LR (prop. odds)	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.02	0.724
Ordinal LR (prop. odds)	Duke MRI (external)	637	0.325	0.520	-1.05	0.92	0.778
Ordinal LR (prop. odds)	Duke MRI (intercept-updated)	637	0.325	0.325	+0.00	0.92	0.777
XGBoost	METABRIC (internal CV)	2386	0.502	0.503	-0.00	1.02	0.755
XGBoost	MSK-2018 (external)	1573	0.676	0.474	+0.98	1.03	0.726
XGBoost	MSK-2018 (intercept-updated)	1573	0.676	0.676	+0.00	1.03	0.726
XGBoost	Duke MRI (external)	637	0.325	0.519	-1.06	0.97	0.789
XGBoost	Duke MRI (intercept-updated)	637	0.325	0.325	-0.00	0.97	0.789

Table S4: Multiclass calibration-in-the-large: observed versus mean predicted prevalence of each grade (%).

Model	Setting	Obs. G1	Pred. G1	Obs. G2	Pred. G2	Obs. G3	Pred. G3
Multinomial LR	METABRIC (internal CV)	8.9	8.9	40.9	40.9	50.2	50.2
Multinomial LR	MSK-2018 (external)	5.7	8.5	26.7	44.3	67.6	47.2
Multinomial LR	Duke MRI (external)	17.7	7.6	49.8	40.7	32.5	51.7
Ordinal LR (prop. odds)	METABRIC (internal CV)	8.9	9.0	40.9	40.9	50.2	50.1
Ordinal LR (prop. odds)	MSK-2018 (external)	5.7	9.1	26.7	43.5	67.6	47.4
Ordinal LR (prop. odds)	Duke MRI (external)	17.7	8.1	49.8	39.9	32.5	52.0
XGBoost	METABRIC (internal CV)	8.9	8.9	40.9	40.8	50.2	50.3
XGBoost	MSK-2018 (external)	5.7	8.7	26.7	43.8	67.6	47.4
XGBoost	Duke MRI (external)	17.7	8.2	49.8	39.9	32.5	51.9

Table S5: Missing data: multiple imputation with pooled predictions and complete-case analysis versus the primary single in-fold imputation (METABRIC, nested cross-validation).

Model	Missing-data method	n	Macro-AUC (95% CI)	G3-AUC	Macro-F1	Brier
Multinomial LR	Multiple imputation (m=20, pooled predictions)	2386	0.724 (0.707–0.742)	0.752	0.465	0.490
Multinomial LR	Complete-case analysis	1800	0.729 (0.709–0.747)	0.761	0.455	0.486
Multinomial LR	Single in-fold imputation (primary)	2386	0.725 (0.707–0.742)	0.752	0.465	0.490
XGBoost	Multiple imputation (m=20, pooled predictions)	2386	0.726 (0.710–0.745)	0.755	0.465	0.489
XGBoost	Complete-case analysis	1800	0.727 (0.709–0.747)	0.760	0.453	0.487
XGBoost	Single in-fold imputation (primary)	2386	0.723 (0.708–0.743)	0.753	0.458	0.492

Table S6: Minimum sample size for developing the multinomial model (Riley criteria applied to each pairwise sub-model).

Sub-model	Share of patients	Anticipated R^2_{CS}	Max R^2_{CS}	n (criterion i)	n (criterion ii)
G1 vs G2	0.498	0.039	0.610	4477	636
G1 vs G3	0.591	0.180	0.572	758	528
G2 vs G3	0.911	0.171	0.747	521	263
Criterion (iii): each grade prevalence ±0.05					385
Required (maximum) / available					4477 / 2386

Table S7: Sensitivity of the Saudi-weighted METABRIC grade distribution (%) to the assumed share of patients aged ≤50 years in the Dammam series (which reports only the median age).

Assumed share ≤50 y	ESS	G1	G2	G3
40%	1278	6.3	33.2	60.5
45%	1187	6.3	33.0	60.7
50%	1088	6.2	32.9	60.9
55%	988	6.2	32.7	61.1
60%	892	6.1	32.6	61.3

Table S8: Ljubljana sensitivity analysis ("degree of malignancy"; age band, menopausal status, tumour-size band, nodes, node capsule invasion; 10×5-fold CV).

Model	n	Macro-AUC (SD over repeats)	Macro-F1	G3-AUC
Majority class	286	0.497 (0.000)	0.208	0.498
Multinomial LR	286	0.636 (0.011)	0.425	0.691
Ordinal LR (prop. odds)	286	0.618 (0.010)	0.397	0.694
Random forest	286	0.626 (0.018)	0.392	0.698
XGBoost	286	0.622 (0.014)	0.402	0.686

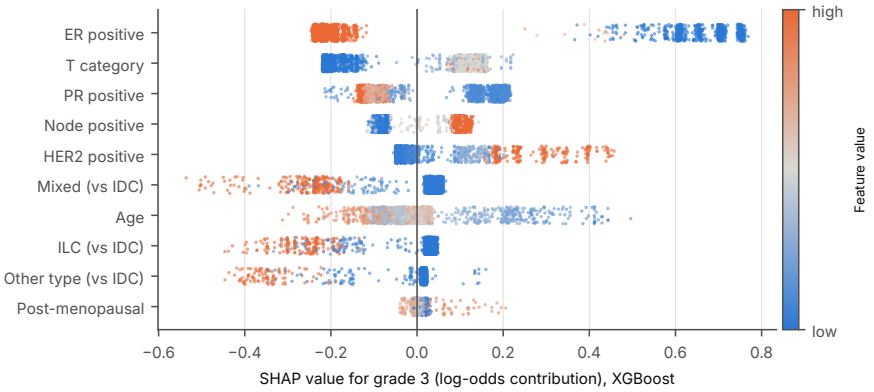


Figure S1: SHAP beeswarm for the grade-3 output of XGBoost in METABRIC; colour encodes the (imputed) feature value.

Table S9: TRIPOD+AI checklist: where each item is reported.

Item	Topic	D/E	Location in this paper
1	Title	D;E	Title: prediction of grade, development and external validation, open data
2	Abstract	D;E	Structured abstract (TRIPOD+AI for Abstracts items)
3a	Background: healthcare context	D;E	Introduction, paragraphs 1 and 3
3b	Background: target population and intended use	D;E	Introduction, paragraph 1 (baseline for image models, registry imputation, quality assurance); Discussion, *Implications*
3c	Background: known biases/inequalities	D;E	Introduction, paragraph 3 (Saudi/Gulf case-mix, absence of open regional data)

Item	Topic	D/E	Location
4	Objectives	D;E	Introduction, *Contributions*
5a	Data sources	D;E	Methods, *Data sources*; Table 1
5b	Dates of data collection	D;E	Methods, *Data sources*; Supplementary Table S1; Data availability (download date and checksums)
6a	Setting and centres	D;E	Methods, *Data sources*
6b	Eligibility criteria	D;E	Methods, *Harmonisation* (rules i–iv)
6c	Treatments received	D;E	Not applicable at diagnosis (grade is a baseline characteristic); Limitations
7	Data preparation	D;E	Methods, *Harmonisation*; repository `harmonise.py`
8a	Outcome definition	D;E	Methods, *Outcome*: invasive Nottingham grade 1–3; grade 3 vs 1–2
8b	Outcome assessment	D;E	Assigned by pathologists at each source; differences discussed under Limitations
8c	Blinding of outcome assessment	D;E	Not applicable (retrospective records); Limitations
9a	Choice of predictors	D	Methods, *Outcome, predictors and feature sets*
9b	Predictor definitions	D;E	Methods, *Harmonisation*; repository code
9c	Blinding of predictor assessment	D;E	Not applicable (routine records); leakage quarantine
10	Sample size	D;E	Methods, *Sample size*; Supplementary Table S6
11	Missing data	D;E	Methods, *Missing data*; Supplementary Table S5
12a	Data use (development/evaluation)	D;E	Methods, *Validation*; Figure 1
12b	Pre-processing	D	Methods, *Model development* (in-fold imputation and scaling)
12c	Model type and building	D	Methods, *Model development*
12d	Heterogeneity across clusters	D;E	LODO validation; cohort-level results (Tables 4, S2)
12e	Performance measures	D;E	Methods, *Performance measures*
12f	Model updating	E	Methods, *Performance measures* (intercept update)
12g	Comparison with other models	D;E	Model ladder; paired Δ vs LR (Table 3)
13	Class imbalance	D;E	Methods, *Model development* (no resampling; prevalence-adjusted rule)

Item	Topic	D/E	Location
14	Fairness	D;E	Results by country/cohort; Saudi transport (Section 2.12)
15	Model output	D	Probabilities for each grade
16	Training versus evaluation differences	D;E	Methods, *Validation*; frozen external models
17	Ethical approval	D;E	Declarations
18a	Funding	D;E	Declarations
18b	Conflicts of interest	D;E	Declarations
18c	Protocol	D;E	Pre-specified analysis plan in `configs/experiment.yaml` (repository)
18d	Registration	D;E	Not registered (secondary analysis of open data); Declarations
18e	Data sharing	D;E	Data availability
18f	Code sharing	D;E	Code availability
19	Patient and public involvement	D;E	Declarations
20a	Participant flow	D;E	Supplementary Table S1
20b	Participant characteristics	D;E	Table 1; Figure 2
20c	Comparison of development and evaluation data	E	Table 1; Results, *Cohorts*
21	Number of participants and outcome events	D;E	Tables 1, 3, 4
22	Full model specification	D	Code and fitted coefficients in the repository; ordinal odds ratios (Table 5)
23a	Performance with CIs	D;E	Tables 3, 4; Figure 4
23b	Performance in subgroups	D;E	By cohort/country; Saudi-weighted performance (Table 8)
24	Model updating results	E	Figure 5; Supplementary Table S3
25	Interpretation	D;E	Discussion, *Principal findings*
26	Limitations	D;E	Discussion, *Strengths and limitations*
27a	Usability: handling poor-quality or missing input	D	Methods, *Missing data*; Discussion

Item	Topic	D/E	Location
27b	Usability: user interaction and expertise	D	Discussion, *Implications*
27c	Next steps and generalisability	D;E	Discussion, *Implications and next steps*